

1. Data Collection

Data collection is the process of gathering relevant data from various sources. The quality of the collected data directly affects the performance of the machine learning model. In this assignment, student performance data is collected and stored in a CSV file. The dataset contains study hours, attendance, assignments, and marks.

2. Data Cleaning

Data cleaning involves checking for missing values, duplicate records, and incorrect data entries. Clean data improves model accuracy and reliability. In this dataset, all values are complete and numerical. Therefore, no major cleaning is required.

3. Feature Selection

Feature selection is the process of choosing the most relevant input variables. Good feature selection helps the model learn meaningful patterns. In this assignment, Study_Hours, Attendance, and Assignments are selected as features. Marks is selected as the target variable.

4. Train-Test Split

The dataset is divided into training and testing sets. The training data is used to train the model, while the testing data is used to evaluate its performance. An 80:20 split is commonly used because it provides enough data for both training and testing.

5. Model Training

Model training is the process of teaching the algorithm using training data. The model learns relationships between features and the target variable. In this assignment, Linear Regression is used for training.

6. Testing

Testing is performed using the unseen test dataset. The trained model makes predictions on test data. This helps determine how well the model can generalize to new data.

7. Evaluation

Evaluation measures the model's performance using metrics such as MAE and R^2 Score. These metrics help determine the accuracy and effectiveness of the model. Better evaluation scores indicate a better model.

8. Prediction

Prediction is the final step where the trained model is used to predict outcomes for new data. In this assignment, the model predicts student marks based on study hours, attendance, and assignments.